

Programming Assignment

(1) Fit the following four points by the cubic splines

i	0	1	2	3
x_i	1	2	3	4
y_i	1	5	11	8

Use the end conditions $y_0'' = y_3'' = 0$. Hence, compute (i) $y(1.5)$ and (ii) $y'(2)$.

(2) Determine the least squares approximation of the type $ax^2 + bx + c$, to the function 2^x at the points $x_i = 0, 1, 2, 3, 4$.